IIIF Document Information Extraction — Weekly Briefing

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Meeting: 11:00–11:20 · **Room:** INN 128 · **Zoom:** https://epfl.zoom.us/j/2693459226

Suggested oral report

Good morning. This week, I studied the main IIIF discovery mechanisms and designed a seven-module harvesting pipeline. I implemented and tested the first three modules: provider census and selection, entry-URL validation, and IIIF resource discovery.

Module 1 currently combines 95 unique candidate records: 69 from the official IIIF Finding Resources guide, 25 from Awesome IIIF, and one from the IIIF Registry. Module 2 checks HTTP status, redirects, response type, elapsed time, JSON parsing, and basic IIIF type. Module 3 discovers resources through Change Discovery streams, IIIF Collections, direct Manifest links, and links declared in HTML pages.

The tested routes produced 21,872 unique Manifest URLs: 21,853 from the Bodleian Change Discovery stream, 18 from Fragmentarium, and one from a Biblissima demonstration. This is a reproducible baseline, not the total available from all 95 candidates.

Testing revealed that the catalogue converter normally retains only the first source-entry link, which is often a homepage. Institutions instead expose IIIF through Change Discovery, Collections, institutional APIs, URL templates, or record-level links. I therefore propose to retain and classify all relevant links in Module 1, validate every candidate endpoint in Module 2, and add Collection pagination plus API/template adapters in Module 3.

Verified evidence

Area	Result
Module 1	95 candidates = 69 official-guide + 25 Awesome IIIF + 1 Registry + 0 manual
Module 2	1 reachable JSON; 65 reachable non-JSON; 1 too large; 12 restricted; 4 temporary failures; 12 unreachable
Module 3	21,872 unique Manifest URLs = Bodleian 21,853 + Fragmentarium 18 + Biblissima demo 1
Tests rerun today	Module 1: 13; Module 2: 11; Module 3: 16 — **40/40 passed**
Interpretation	Candidates are not confirmed APIs; HTML or one failed URL does not prove that an institution lacks IIIF

Problem and correction

IIIF has no mandatory universal discovery endpoint. The Registry is voluntary and incomplete; the official guide mainly explains record-level Manifest discovery; Awesome IIIF mixes providers, viewers, software, Presentation API and Image API implementations. Current coverage therefore measures input and adapter limitations rather than worldwide IIIF availability.

- **Module 1:** preserve the raw entry and every relevant link; classify homepage, Change Discovery, Collection, API documentation/search, Manifest examples and URL templates; distinguish providers from viewers, software and Image-only services.
- **Module 2:** check every candidate link and retain status, redirect, final URL, content type, timing, JSON/type evidence and failures; do not stop after one success.
- **Module 3:** add Presentation 2/3 Collection **next** pagination, API/template adapters, loop detection, checkpoints, caching, limits, rate control, retry/backoff and resumable output.

Use bounded samples before complete harvests. BSB advertised about 3.2 million Collection members; Carnegie exposed about 908 Manifests. An Image API **info.json** must not be counted as a Presentation Manifest.

Decisions and next actions

Confirm: revised architecture; first complete provider; rights allowlist; treatment of Image-only records; common schema and handoff; server resources and rate limits.

Then: fix Module 1 multi-link ingestion and tests; extend Module 2; add Collection/API/template support to Module 3; sample BSB and Carnegie; implement Module 4 Manifest validation; continue with rights, metadata and common-schema export.

IIIF Standard APIs — Meaning and Use

General request method

IIIF resources normally use HTTP(S) and JSON or JSON-LD. Follow redirects, set an Accept header, record the final URL/status/content type, and check both context and resource type.

```
curl.exe -L -H "Accept: application/ld+json, application/json" URL -o
↪  out.json
python -m json.tool out.json > pretty.json
```

Presentation API 3.0

Purpose: describes how a compound digital object should be presented. A **Manifest** normally represents one object; its **Canvases** represent ordered views, pages, surfaces or time spans. A **Collection** groups Manifests and child Collections.

How to use it:

1. Request a known Manifest or Collection URL.
2. Verify the IIIF context and **type**.
3. For Presentation 3, read Collection/Manifest **items**. In Presentation 2, Collections commonly use **manifests/collections**, while Manifests commonly use **sequences**.
4. Follow Collection **next** pagination and nested Collections.
5. Preserve **id**, label, metadata, rights, requiredStatement, provider and source URL. Human-readable metadata is not guaranteed to be semantically normalized.

```
curl.exe -L "MANIFEST_OR_COLLECTION_URL" -o presentation.json
```

Official specification: https://iiif.io/api/presentation/3.0/

Image API 3.0

Purpose: delivers image information and derivatives. It can crop, resize, rotate and select quality/format. It does not describe the complete cultural object.

How to use it:

```
GET {image-service-id}/info.json
GET {image-service-id}/{region}/{size}/{rotation}/{quality}.{format}

# Example
GET https://example.org/image-service/id/full/max/0/default.jpg
```

Read **info.json** to learn dimensions, profile and supported features. Construct image requests in the required order: region, size, rotation, quality and format. Do not count info.json as a Manifest.

Official specification: https://iiif.io/api/image/3.0/

Change Discovery API 1.0

Purpose: publishes additions, updates, deletions and moves of IIIF resources so a client can synchronize incrementally.

How to use it:

1. Fetch the provider's **OrderedCollection**.
2. Open its first/last page and follow **next/prev** as appropriate.
3. Read every page's **orderedItems**.
4. For each Activity, inspect **type**, **object.id**, **object.type** and **endTime**.
5. Retrieve current Manifest/Collection objects for Create/Update; preserve or remove state for Delete/Remove/Move.
6. Store the last processed activity/time for later incremental runs.

```
curl.exe -L "https://iiif.bodleian.ox.ac.uk/iiif/activity/all-changes"
↪  -o activity.json
```

The Activity reports that a resource changed; it does not necessarily contain a field-level diff.

Official specification: https://iiif.io/api/discovery/1.0/

Content Search API 2.0

Purpose: searches annotations associated with a Manifest or Collection, commonly OCR/transcription text. It is search **inside known IIIF resources**, not a universal provider-discovery API.

How to use it:

1. Find a declared SearchService2 in the Presentation resource.
2. send its search endpoint a URL-encoded **q** parameter;
3. read the returned AnnotationPage/AnnotationCollection and **items**;
4. follow **next** pages and use Annotation targets to locate the matching Canvas or region.

```
GET https://example.org/service/manifest/search?q=bird
```

Official specification: https://iiif.io/api/search/2.0/

Registry, Collections and templates

The Registry is a directory of voluntarily registered streams, not a complete IIIF census:

```
curl.exe -L "https://registry.iiif.io/index.json" -o registry.json
```

Verified Collection examples:

```
curl.exe -L "https://api.digitale-
↪  sammlungen.de/iiif/presentation/v2/collection/top" -o
↪  bsb.json
curl.exe -L
↪  "https://cmoa-records-images.s3.amazonaws.com/collection/top.json"
↪  -o cmoa.json
```

Some providers require non-IIIF enumeration before Manifest retrieval: Europeana and Harvard use keyed institutional APIs; Gallica and Internet Archive expose search identifiers that can be inserted into deterministic Manifest URL templates. These adapters belong in Module 3. Credentials must stay outside Git.

Pipeline Diagrams

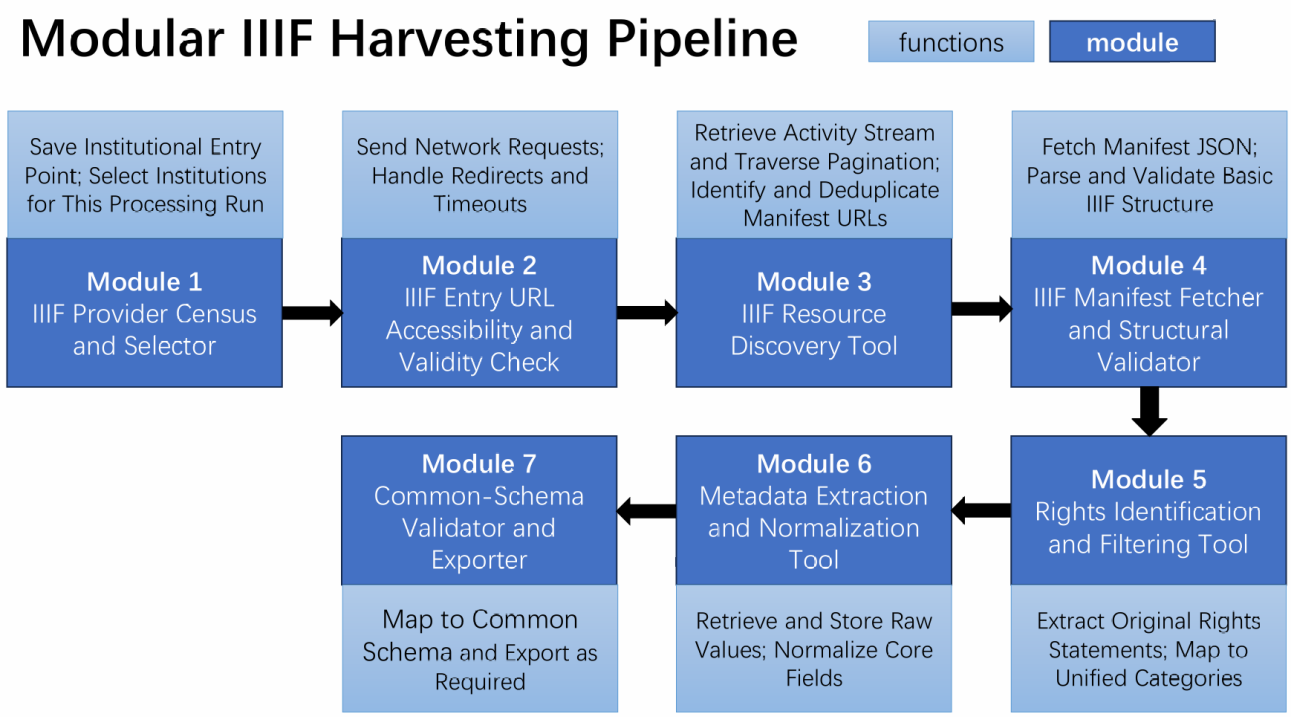


Figure 1. Seven-module IIIF harvesting pipeline

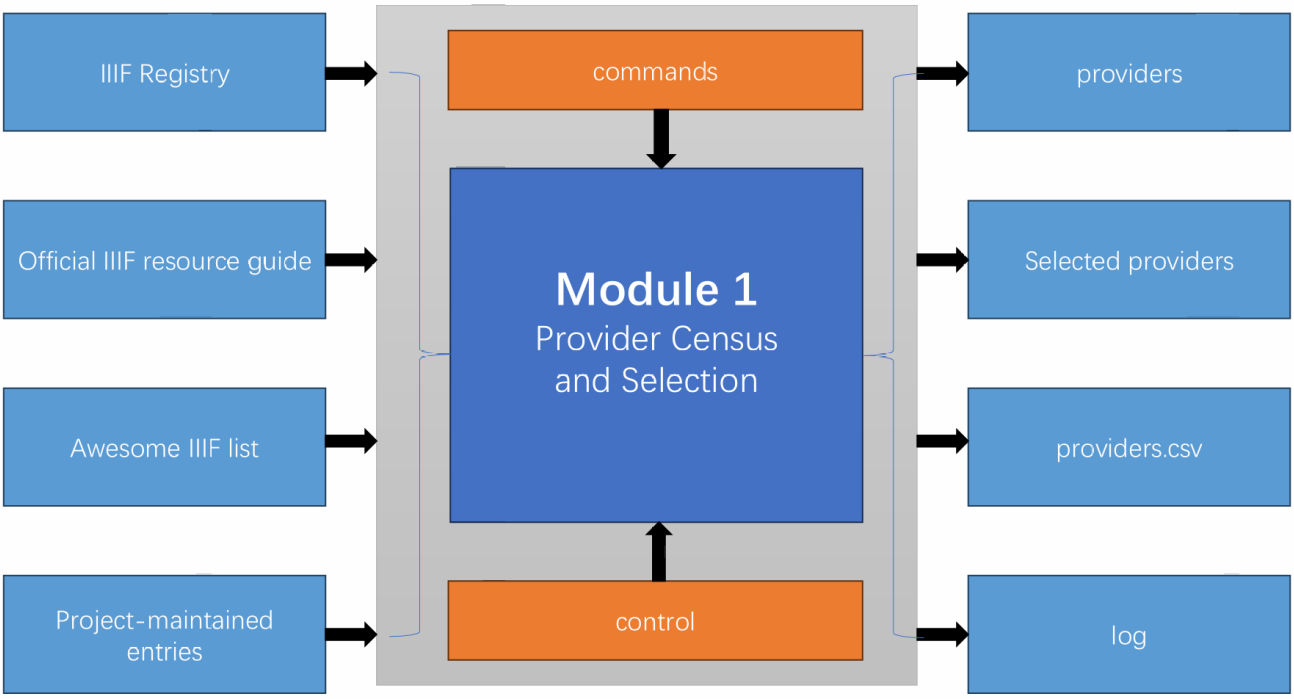


Figure 2. Module 1 inputs, control and outputs